Homework 08

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Question 1: Boundary and margin

(a) Distances and functional margins

$$f(x) = 0.8x_1 + 0.6x_2, \qquad \|(0.8, 0.6)\|_2 = 1, \qquad d(x, \{f = 0\}) = \frac{|f(x)|}{\|(0.8, 0.6)\|_2} = |f(x)|.$$

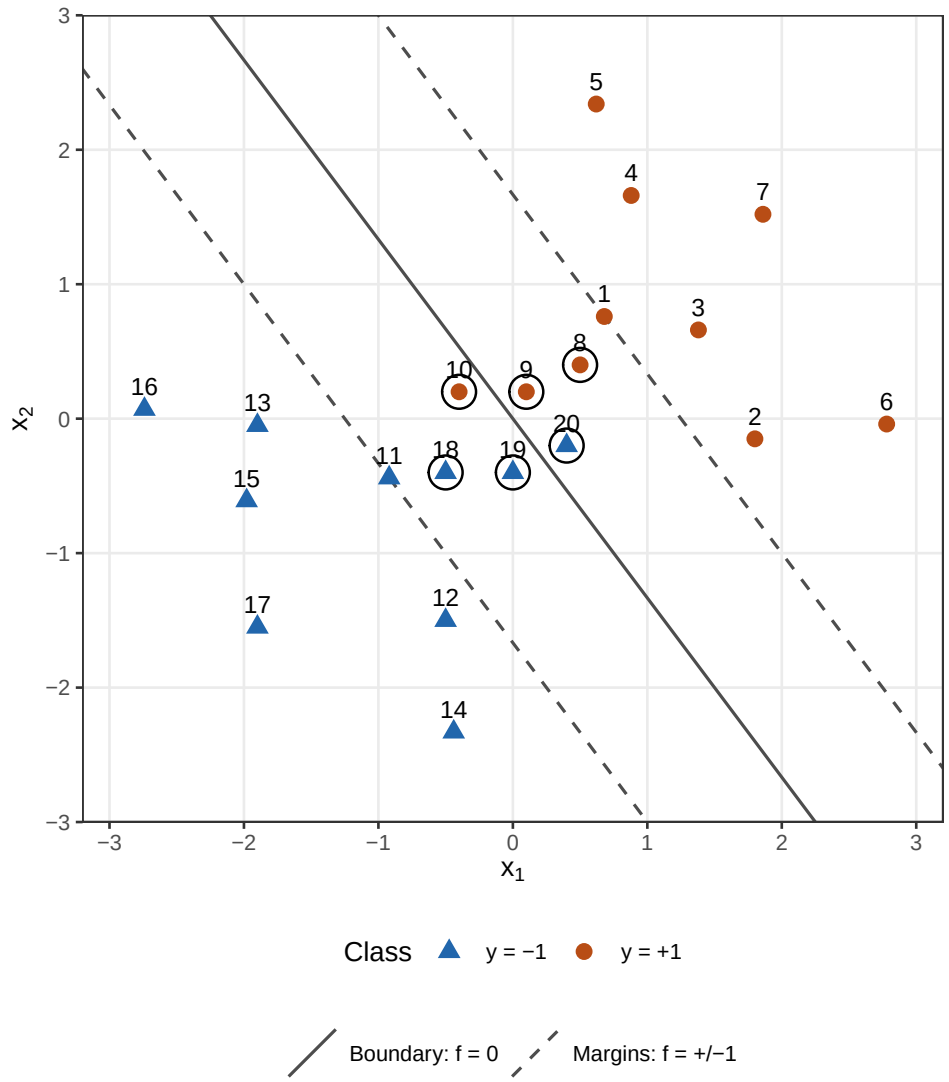
$$m_i = y_i f(x_i), \qquad \begin{cases} m_i < 1 & : \text{inside}, \\ m_i = 1 & : \text{on}, \\ m_i > 1 & : \text{beyond}. \end{cases}$$

Table 1: Distance to the boundary and class-specific margin position.

Obs.	x_1	x_2	y	$f(x)$	Distance	$yf(x)$	Position
1	0.68	0.76	1	1.00	1.00	1.00	On
2	1.80	-0.15	1	1.35	1.35	1.35	Beyond
3	1.38	0.66	1	1.50	1.50	1.50	Beyond
4	0.88	1.66	1	1.70	1.70	1.70	Beyond
5	0.62	2.34	1	1.90	1.90	1.90	Beyond
6	2.78	-0.04	1	2.20	2.20	2.20	Beyond
7	1.86	1.52	1	2.40	2.40	2.40	Beyond
8	0.50	0.40	1	0.64	0.64	0.64	Inside
9	0.10	0.20	1	0.20	0.20	0.20	Inside
10	-0.40	0.20	1	-0.20	0.20	-0.20	Inside
11	-0.92	-0.44	-1	-1.00	1.00	1.00	On
12	-0.50	-1.50	-1	-1.30	1.30	1.30	Beyond
13	-1.90	-0.05	-1	-1.55	1.55	1.55	Beyond
14	-0.44	-2.33	-1	-1.75	1.75	1.75	Beyond
15	-1.98	-0.61	-1	-1.95	1.95	1.95	Beyond
16	-2.74	0.07	-1	-2.15	2.15	2.15	Beyond
17	-1.90	-1.55	-1	-2.45	2.45	2.45	Beyond
18	-0.50	-0.40	-1	-0.64	0.64	0.64	Inside
19	0.00	-0.40	-1	-0.24	0.24	0.24	Inside
20	0.40	-0.20	-1	0.20	0.20	-0.20	Inside

I used $|f(x)|$ to calculate each point's distance from the boundary because the coefficient vector has length 1. Then I compared $yf(x)$ with 1. A point is inside the margin if the value is less than 1, on the margin if it equals 1, and outside if it is greater than 1. The table shows all the results.

(b) Boundary and margin plot



Labels are observation numbers. Black circles identify observations inside their class margin.

The solid line is the boundary, and the two dashed lines are the margins. Different colors show the two classes, and the numbers are the IDs from the original data. The six circled points are inside the margin because their $yf(x)$ values are less than 1.

(c) Counts and misclassified observations

Inside	On	Beyond
6	2	12

Misclassified observations: 10, 20; $\quad y_{10}f(x_{10}) = y_{20}f(x_{20}) = -0.20.$

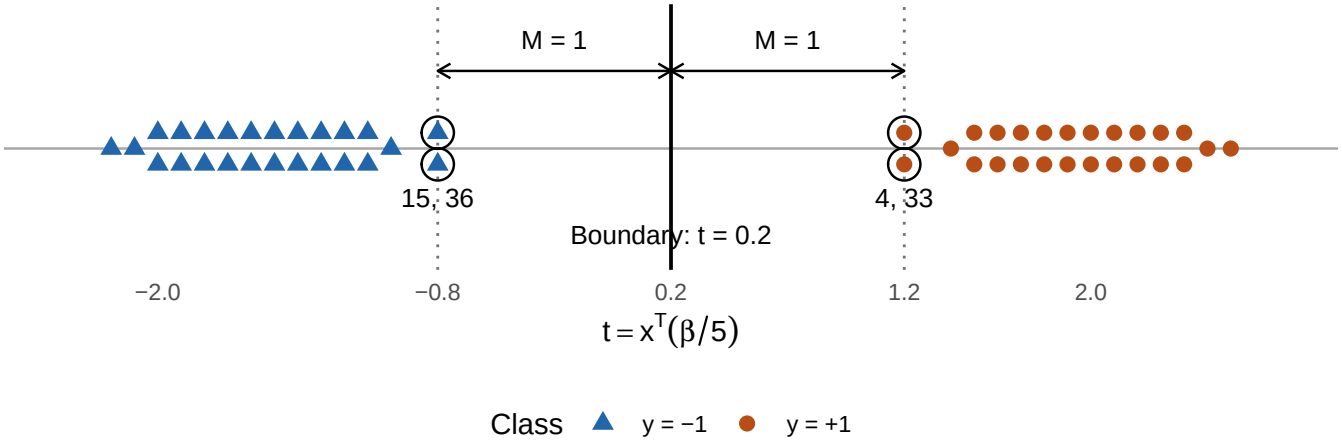
I found 6 points inside the margin, 2 on it, and 12 outside. Points 10 and 20 are misclassified because they fall on the other class's side of the boundary. Both have $yf(x) = -0.2$. The other four points inside the margin are close to the boundary, but they are classified correctly.

Question 2: Margin in four dimensions

(a) Projection and geometric margin

$$\beta = (1, 2, -2, 4)^\mathsf{T}, \qquad \|\beta\|_2 = 5, \qquad \mathbf{u} = \frac{\beta}{5} = (0.2, 0.4, -0.4, 0.8)^\mathsf{T}.$$

$$t_i = \mathbf{x}_i^\mathsf{T}\mathbf{u}, \qquad f(\mathbf{x}_i) = -1 + 5t_i, \qquad t_0 = \frac{1}{5} = 0.2, \qquad M = \min_i |t_i - t_0| = \frac{\min_i |f(\mathbf{x}_i)|}{5} = 1.$$



Vertical offsets separate coincident projections. Circled points are closest to the boundary.

Table 2: The four observations closest to the decision boundary.

Obs.	y	t_i	$f(x_i)$	$y_i f(x_i)$	Distance
4	1	1.2	5	5	1
15	-1	-0.8	-5	5	1
33	1	1.2	5	5	1
36	-1	-0.8	-5	5	1

The length of β is 5, so I first divided it by 5 and projected the points onto that direction. The boundary is at 0.2, and the closest points are 4, 15, 33, and 36. Their distances from the boundary are all 1, so $M = 1$.

(b) SVM normalization

$$c = \min_i y_i f(\mathbf{x}_i) = 5, \qquad \tilde{f}(\mathbf{x}) = \frac{f(\mathbf{x})}{5}, \qquad \tilde{\beta} = (0.2, 0.4, -0.4, 0.8)^\mathsf{T}, \quad \tilde{\beta}_0 = -0.2.$$

$$\min_i y_i \tilde{f}(\mathbf{x}_i) = 1, \qquad \|\tilde{\beta}\|_2 = 1, \qquad M = \frac{1}{\|\tilde{\beta}\|_2} = 1.$$

The smallest functional margin is 5, so I divided all the original coefficients and the intercept by 5. The new coefficients are $(0.2, 0.4, -0.4, 0.8)$, and the intercept is -0.2. This makes the smallest $yf(x)$ equal to 1. Since everything is divided by the same positive number, the boundary and the classification results stay the same.

Question 3: Peer review of a Homework 7 skill

Assigned repository: tewarysheetal/stat432-fall2026.

Skill source: the complete `SKILL.md` on pages 7-8 of the assigned Homework 7 report.

Skill run record

Item	Value
Random seed	432
Train / test size	369 / 93
Validation	5 stratified folds
Candidates	Ordinary LDA and QDA
Selected model	LDA
Held-out test error	19/93 = 20.43%

Model	Mean CV error
LDA	26.57%
QDA	32.26%

Run completed without an early-stop steer.

The one-page report and reproducible code are saved separately.

Overall assessment

I think the steps in this skill are clear, and the report is easy to understand. LDA was selected in the end, with a test error rate of 20.43%. I think the following three areas could be improved.

Major item 1

I noticed that the original report tried 36 settings, while this run only compared LDA and QDA. The data splits were also different. The skill allows this, but it makes the results harder to compare directly. I think adding default settings and clearly stating which settings were actually used would make it easier for others to repeat the analysis.

Major item 2

Your skill says to include the code, but I could not seem to find it in the PDF. This means I cannot reproduce the results in the original report. Including the code that was actually used would be helpful next time.

Major item 3

I noticed that 13 of the 32 people who actually had the disease were misclassified in this run. This shows that an overall error rate of 20.43% does not mean both classes are predicted well. The original report also mentioned missed cases. Maybe there is a way to improve this and reduce the number of people with the disease who are misclassified.

Peer review GitHub Issue: tewarysheetal/stat432-fall2026#4.